

Smart Traffic Lights Using Local Edge AI: Architecture, Limitations, and Prioritization Protocols

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Abstract—Smart traffic-light systems are being actively researched to reduce vehicle waiting times and dynamically respond to fluctuating road conditions. Traditional fixed-time controllers are simple and cost-effective but fail during unexpected congestion. While cloud-based AI solutions offer high computational capacity, they introduce network latency and dependency on continuous internet availability. Local edge processing presents a balanced alternative by executing vision tasks near the roadside. This paper reviews current literature on traffic signal control, critically evaluates micro-controller processing capabilities, proposes a real-time system architecture, and identifies key gaps regarding emergency vehicle routing.

Index Terms—Traffic Signal Control, Edge AI, Computer Vision, Local Processing, Intelligent Transportation Systems (ITS)

I. INTRODUCTION & LITERATURE REVIEW

Adaptive traffic control aims to mitigate urban congestion by continuously modifying signal timing based on real-time vehicle density [1]. Conventional fixed-time traffic signals operate on predefined clock cycles. While inexpensive, these static configurations cannot dynamically adapt when unexpected bottlenecking occurs.

Recent advances employ computer vision and artificial intelligence (AI) to estimate vehicle queue lengths. Instead of switching signals solely on static timers, systems utilize dynamic road metrics. Comprehensive reviews demonstrate a expanding focus on deep reinforcement learning and machine vision for adaptive signal regulation [1], [2].

TABLE I
COMPARISON OF TRAFFIC SYSTEM ARCHITECTURES

System Type	Decision Latency	Data Usage	Setup Cost
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Systems	2.0 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

Centralized cloud AI architectures aggregate raw camera video streams at distant server nodes. Despite providing vast compute resources, cloud dependency introduces non-trivial network latency and bandwidth overheads. Conversely, local smart-camera systems deploy edge computation units near

the signal hardware, significantly limiting bandwidth demands while accelerating decision responsiveness, as outlined in Table I.

A. Critical Evaluation of Low-Cost Microcontrollers

Literature occasionally features questionable operational claims. For instance, Smith et al. [3] alleged that an 8-bit Arduino Uno processed 10,000 live HD traffic video streams concurrently with zero latency. From an engineering standpoint, an 8-bit microcontroller operating at 16 MHz with 2 KB SRAM lacks the memory bandwidth, frame-buffer capacity, and computational throughput required to decode or analyze even a single high-definition stream in real time. Students and researchers must critically assess hardware capabilities prior to adopting literature assertions.

II. PROPOSED SYSTEM ARCHITECTURE

The proposed intelligent traffic framework isolates localized latency-sensitive operations from non-critical global logging. Figure 1 outlines the operational sequence across four primary system modules.

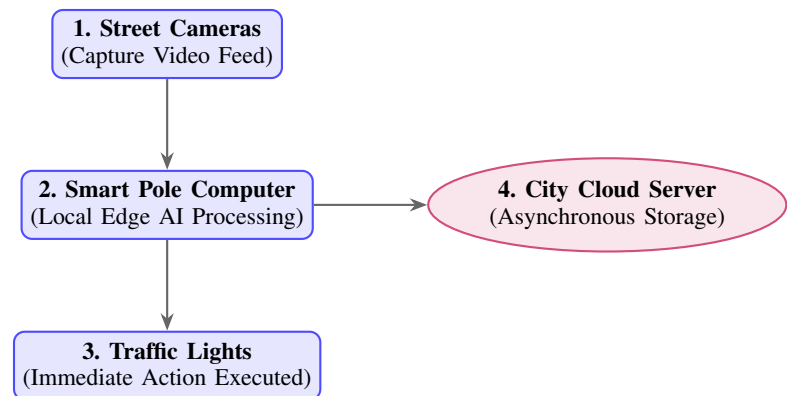


Fig. 1. Data and control flow diagram of the Local AI Smart Traffic Architecture.

The functional steps are defined as follows:

- 1) **Street Cameras:** Continuous live traffic acquisition focused on active lane segments.

- 2) **Smart Pole Computers:** Roadside edge processing nodes executing light-weight vehicle detection models to compute optimal signal timings.
- 3) **Traffic Lights:** Direct execution targets receiving immediate timing updates from the local pole computer.
- 4) **City Cloud Server:** Secondary destination receiving low-bandwidth summarized metrics asynchronously for long-term analytics and city-wide monitoring.

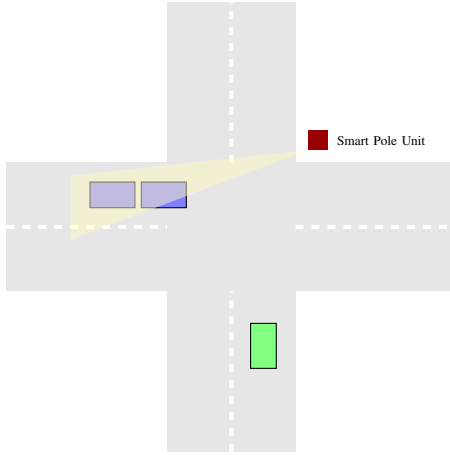


Fig. 2. Visual representation of local roadside computer monitoring vehicle density at an intersection.

III. RESEARCH GAP & FUTURE DIRECTIONS

While current adaptive models effectively balance daily predictable traffic loads, a critical gap remains in handling emergency priority routing. Most existing edge architectures lack automated protocols to detect approaching emergency vehicles (e.g., ambulances) and dynamically establish coordinated signal green corridors. Future work will focus on integrating acoustic and vision-based emergency detection directly into the roadside edge processing workflow.

IV. CONCLUSION

Local AI traffic control strikes a balance between static timed lights and network-dependent cloud systems. By localizing compute operations to Smart Pole units, real-time response times are minimized while maintaining continuous operation during internet outages.

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